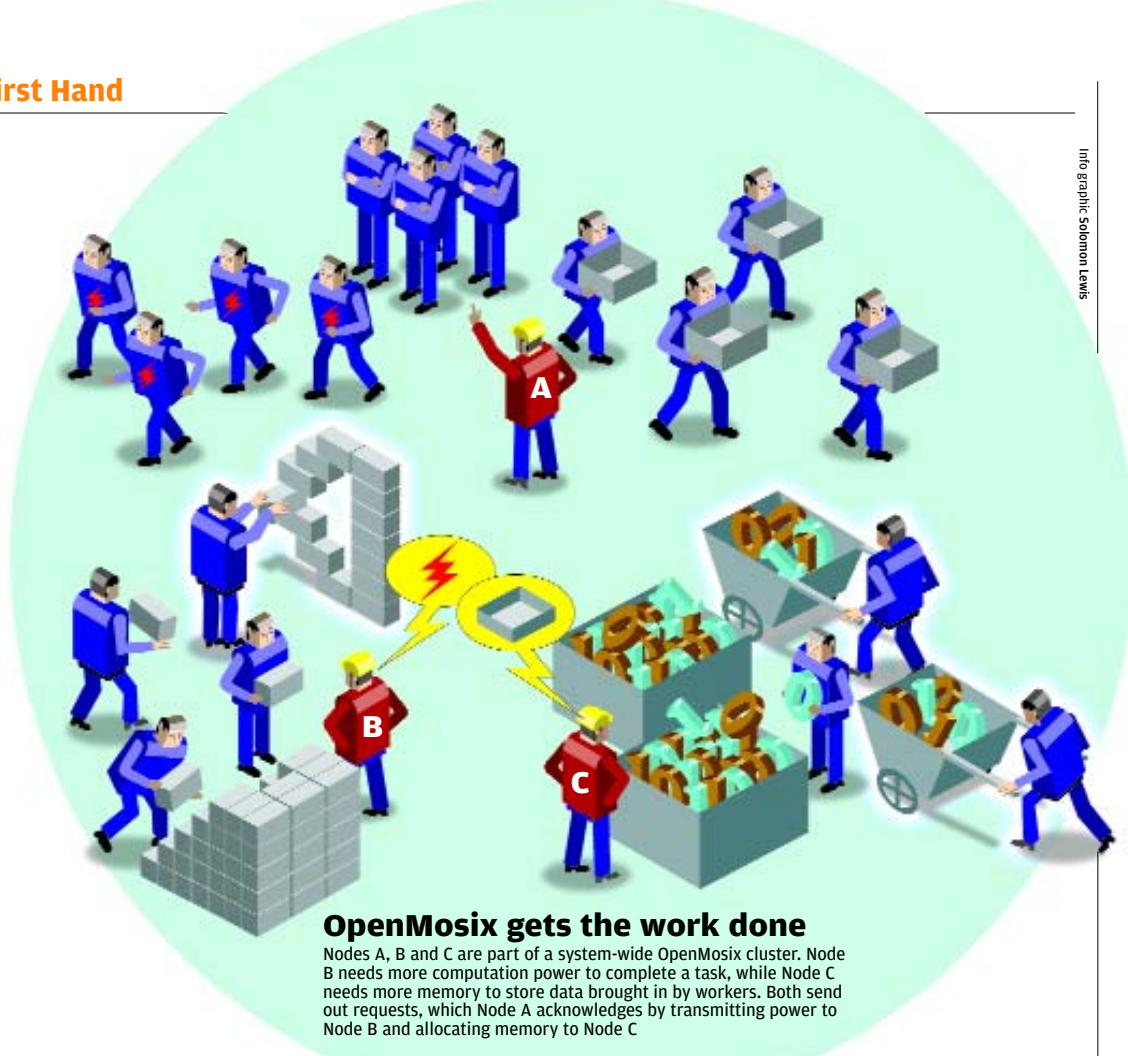


When three young and unassuming geniuses from Jalandhar get together, this is what they come up with—a Linux-based supercomputer! Meet...



OpenMosix gets the work done

Nodes A, B and C are part of a system-wide OpenMosix cluster. Node B needs more computation power to complete a task, while Node C needs more memory to store data brought in by workers. Both send out requests, which Node A acknowledges by transmitting power to Node B and allocating memory to Node C.

Team Prerna

Ahmed Shaikh

There are geniuses amongst us. Young, unassuming characters, destined to change the way you think, work and live.

Abhinav Dhall, 19 years old: The World to a religious homemaker, brother to a fellow student, and son to Tarsem Lal Dhall, an Executive Engineer under the Pushpa Gujral Science City project and a role-model to Abhinav. His favourite movies are *Armageddon* and *Kal Ho Na Ho*. He also enjoys the Backstreet Boys.

Vibhore Jain, 20 years old: Best friend to his sister, Sonu to his mom and his father, an insurance advisor with TATA AIG. He loves classics—*Ben-Hur* is his favourite movie, Mark Twain one of his favourite authors, with Robert Frost's

seminal beauty *Stopping by Woods on a Snowy Evening* his inspiration.

Both Abhinav and Vibhore are second-year students at the D A V Institute of Engineering & Technology (DAVIET), Jalandhar. Under the auspices of Sandeep

Kath, a 24-year-old computer science BTech, and faculty member, the dynamic duo has created a supercomputer for all of us. Codename: Prerna.

How it All Began

DAVIET dabbles with fibre-optics and electronic simulations. Research needs that require computational power upwards of the gigaflop range. Affordable faster machines were needed. What better way than to maximise the potential of the desktop computers that filled two labs within the DAVIET campus? Enter clusters!

A cluster is a collective of PCs or "nodes" with task-masters, which divide tasks between the nodes—the more nodes there are, the faster the cluster. Prerna hired openMosix as the overseer, which sits atop a Linux kernel, openMosix and distributes work amongst the lab PCs.

Each node handles smaller tasks, has greater access to a cluster-wide memory, and finishes a job faster as a single unit. Thus was Prerna born.

"The idea struck us when I read an article on Beowulf in *Digit*," recalls Abhinav. "We began by analysing the technologies available for parallelising computational tasks. We started with PVM on Windows and then tested MPI on Fedora—the most demanding part was that for



Abhinav Dhall, 19
Second year student

Vibhore Jain, 20
Second year student

Sandeep Kath, 24
Faculty Member